

HUBUNGAN POLA PENGGUNAAN DAN JENIS MODE TRANSPORTASI TERHADAP PERSENTASE PENGGUNA DI MASYARAKAT

KELOMPOK 27 :

41425064 YOSEVYN SIPAHUTAR

41425081 LUKAS SIMATUPANG

41425083 INEZ CECILIA TIURMA

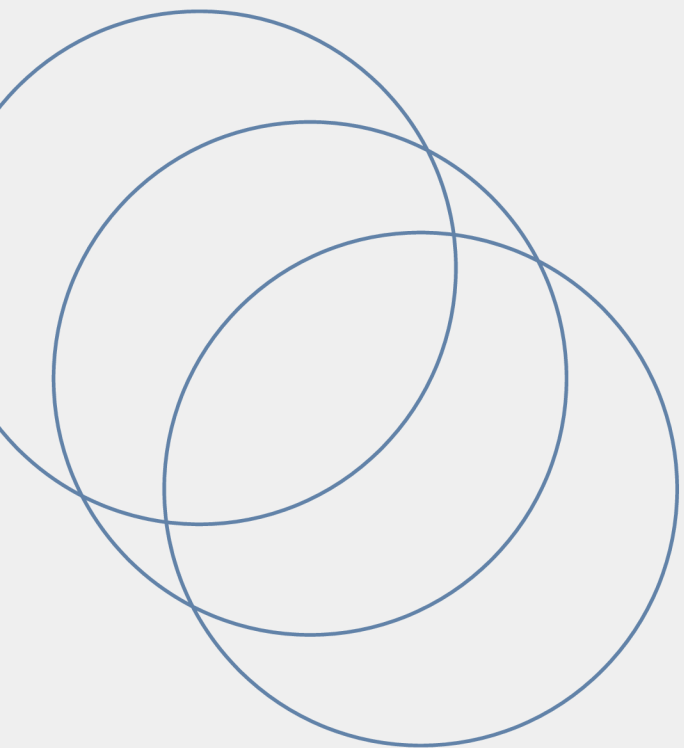
Latar Belakang

01. Transportasi penting bagi mobilitas dan pembangunan sosial-ekonomi

02. Urbanisasi mendorong kebutuhan sistem transportasi yang efisien dan berkelanjutan.

03. Analisis data membantu memahami pola penggunaan moda transportasi antar kelompok.

03. Analisis data membantu memahami pola penggunaan moda transportasi antar kelompok.





Rumusan Masalah

- a. Mode transportasi apa yang memiliki rata-rata percent terbesar?
- b. Apakah terdapat perbedaan signifikan percent antar mode transportasi?



Tujuan

- Memberikan pemahaman mendalam tentang pola dan perbedaan penggunaan moda transportasi di masyarakat.
- Mengetahui distribusi penggunaan moda melalui analisis deskriptif berdasarkan wilayah dan kondisi pengguna.
- Menilai perbedaan signifikan antar kelompok moda dengan uji statistik Kruskal–Wallis.
- Mendukung perencanaan transportasi yang efisien dan berkelanjutan berbasis data.



Data Collection

- Dataset diambil dari platform terbuka Kaggle, berjudul Mode of Transportation Dataset.
- Berisi data pola penggunaan moda transportasi berdasarkan wilayah, etnis, dan tahun.
- Terdiri dari 24 kolom dan 202.203 baris data survei.
- Atribut utama: mode_name, percent, pop_total, reportyear, dan variabel demografis lainnya.
- Digunakan untuk analisis distribusi dan perbedaan penggunaan antar kelompok moda.



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[1]
✓ 5 d

```
import kagglehub

# Download latest version
path = kagglehub.dataset_download("amitzala/transportation-dataset")

print("Path to dataset files:", path)
```

Using Colab cache for faster access to the 'transportation-dataset' dataset.
Path to dataset files: /kaggle/input/transportation-dataset

[4]
✓ 1 d

```
import pandas as pd

# Try reading the CSV with a different encoding
try:
    df = pd.read_csv(path + "/transportation-to-work-1/mode_of_transportation.csv", encoding='latin1')
except UnicodeDecodeError:
    # If latin1 fails, try another common encoding
    df = pd.read_csv(path + "/transportation-to-work-1/mode_of_transportation.csv", encoding='cp1252')

df.head()
```

/tmp/ipython-input-1313982699.py:5: DtypeWarning: Columns (0,7,10) have mixed types. Specify dtype option on import or set low_memory=False.

```
df = pd.read_csv(path + "/transportation-to-work-1/mode_of_transportation.csv", encoding='latin1')
```

	ind_id	ind_definition	reportyear	race_eth_code	race_eth_name	geotype	geotypevalue	geoname	county_name	county_fips	...	pop_total	pop_m
0	42	Percent of residents mode of transportation to...	2000	1.0	AIAN	CA	6.0	California	NaN	NaN	...	73560.0	241

Data ini diambil dari platform Kaggle:

<https://www.kaggle.com/datasets/amitzala/transportation-dataset>

data diambil dari file csv yang bernama

“mode_of_transportation.csv”



Data Overview

- Contoh Kolom Utama:
- mode_name: Jenis moda transportasi
- percent: Persentase pengguna
- pop_total: Jumlah populasi
- reportyear: Tahun laporan
- race_eth_name: Nama kelompok etnis
- geoname: Nama wilayah geografis



Data Understanding

- Contoh Kolom Utama:
- mode_name: Jenis moda transportasi
- percent: Persentase pengguna
- pop_total: Jumlah populasi
- reportyear: Tahun laporan
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Data Understanding

	ind_id	ind_definition	reportyear	race_eth_code	race_eth_name	geotype	geotypevalue	geoname	county_name	county_fips	...	pop_total	pop_mode	percent	LL95CI_percent	UL95CI_percent	percent_se	percent_rse	CA_decile	
0	42	Percent of residents mode of transportation to...	2000	1.0	AIAN	CA	6.0	California	California	6.0	...	73560.0	2414.0	3.281675	NaN	NaN	NaN	NaN	NaN	0.0
1	42	Percent of residents mode of transportation to...	2000	2.0	Asian	CA	6.0	California	California	6.0	...	1634923.0	42902.0	2.624099	NaN	NaN	NaN	NaN	NaN	0.0
2	42	Percent of residents mode of transportation to...	2000	3.0	AfricanAm	CA	6.0	California	California	6.0	...	797818.0	21348.0	2.675798	NaN	NaN	NaN	NaN	NaN	0.0
3	42	Percent of residents mode of transportation to...	2000	4.0	Latino	CA	6.0	California	California	6.0	...	3865125.0	75246.0	1.946793	NaN	NaN	NaN	NaN	NaN	0.0
4	42	Percent of residents mode of transportation to...	2000	5.0	NHOPI	CA	6.0	California	California	6.0	...	42880.0	1088.0	2.537313	NaN	NaN	NaN	NaN	NaN	0.0
5 rows x 24 columns																				



Data Understanding

```
[2]
import pandas as pd

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4	42	Percent of residents mode of transportation to...	2000	5.0	Hispanic	CA	6.0	California	NaN	NaN	...	4000.0	4000.0	0.000000	NaN	NaN



Data Understanding



Imutabilitas Data Geografis: Validasi county_name dan county_fips Berdasarkan mpo_county_list

```
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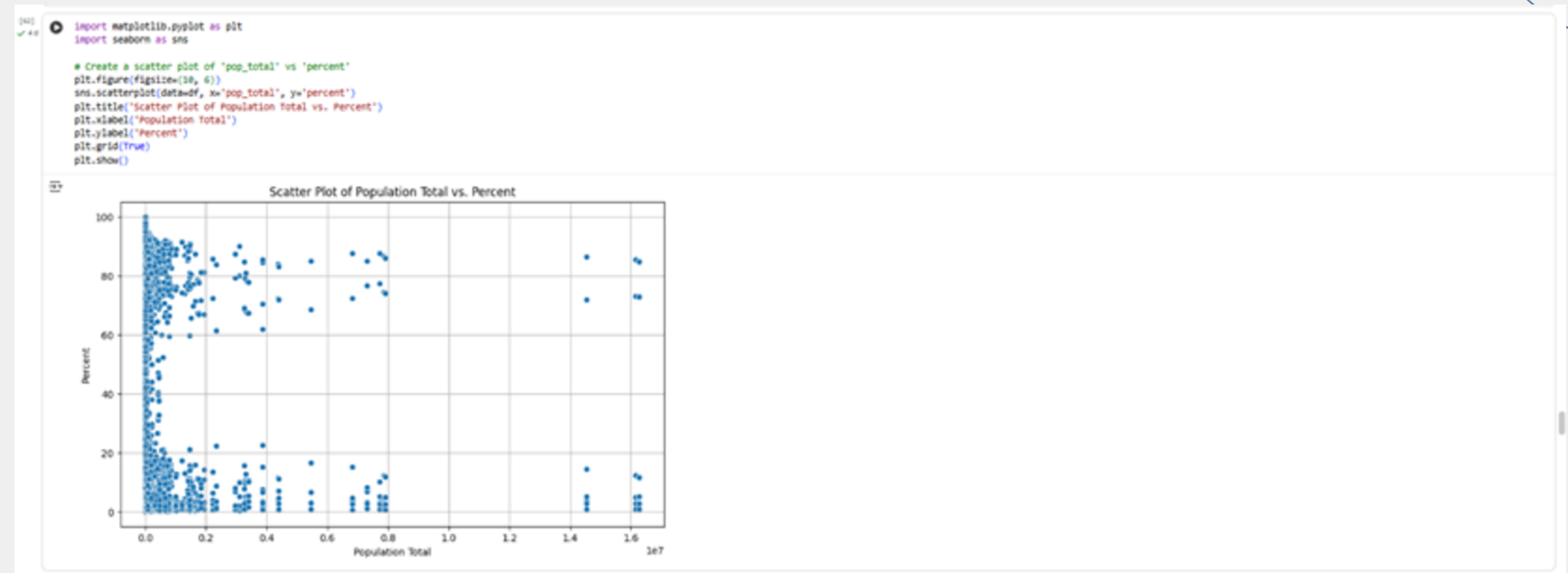
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Analisis korelasi tabel percent dengan pop_total



Menguji dengan: spearman dan pearson

Reasoning: Calculate the Spearman correlation coefficient between 'pop_total' and 'percent' columns to measure their monotonic relationship.

```
[63] ✓ 0 d  correlation_spearman = df['pop_total'].corr(df['percent'], method='spearman')  
print(f"Spearman correlation between pop_total and percent: {correlation_spearman}")  
  
⇒ Spearman correlation between pop_total and percent: 0.04099187824135656
```

Reasoning: Calculate the Pearson correlation coefficient between 'pop_total' and 'percent' columns to measure their linear relationship.

```
[61] ✓ 0 d correlation = df['pop_total'].corr(df['percent'], method='pearson')  
print(f"Pearson correlation between pop_total and percent: {correlation}")  
  
Pearson correlation between pop_total and percent: -0.0001221873790573918
```

